RunCPM 7xtgv-terminal User Guide

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Use this guide for first experiments in the browser and everyday work on physical modules or native Linux/Windows desktops.

CPU speed: `SPEED 65535` selects the fastest throttled ("slow") setting; `SPEED 0` restores unthrottled execution, the default. Smaller nonzero values slow the emulator further. A reported i7-10710U (1.1 GHz) test gave roughly 44 MHz in VTCPU with `SPEED 65535`, on both desktop and web. That is an observed equivalent Z80 benchmark rate, not a fixed MHz setting. See the SPEED reference.

- **Browser:** click the display to type. **Screen text** gives a frozen copy of the complete logical text screen; it is not a live native mirror.
- **Native mirror:** keyboard input and the complete logical CP/M screen for physical hardware or SDL desktop sessions.
- **TFT, SDL or browser canvas:** the board-sized viewport. Glyph zoom changes cell size; pan moves the viewport. Browser **Display** scale only enlarges the canvas presentation. None of these moves CP/M files.
- **CP/M drive/user (DU):** where guest commands read and write files.
- **Native transfer panel's local DU:** a host-card folder, not a command to change the guest's current area.

For when these behaviors and commands were added, see the Features Timeline.

1. Getting started

Start a browser session

Open 7xtgv-terminal in your browser. No checkout, compiler or external terminal is required. The first visit may reload once before the emulator starts.

a. Wait for the CP/M prompt, then click the display to type; a green outline indicates keyboard focus. Press Enter after each command below.

b. Type `VTINFO` to see the selected target and terminal geometry.

c. Type `7XF`, then `LS /`: the prompt uses catalog names and the listing shows the card's drives/areas. `LS` lists the current area, `LL` adds details; see exploring CP/M for navigation.

d. Try palettes or the xterm256 graphics demonstration.

e. Save work in the guest application, use **Stop**, wait for **Stopped cleanly**, then **Download card ZIP** to retain the card's files, settings and history.

The middle **Guides** column offers PDF/ADOC links in guide order and a compact history, browse and change-area reference. ADOC opens the rendered guide on GitHub; the note below the list links Asciidoctor.js for Raw pages and our shared CSS/JS. It is visible by default; **Hide guides** in Controls gives the display more room, and **Show guides** brings it back. An unavailable PDF badge means that guide has not been printed yet.

Choose **Target**, then **Switch target** to try another display. Switching restarts RunCPM with the current memory card: save application work first. It does not flash or connect to a physical board. Smaller displays may show only part of the logical screen; use pan or **Screen text** to inspect it.

Display scale changes only the canvas's on-page size. Glyph zoom/pan and logical terminal geometry are separate controls; see terminal size. **Screen text** can be refreshed, selected or downloaded; **Copy image** and **Download PNG** capture the panel graphics.

Click the display or **Keyboard → display** to resume typing after using page controls. Click or tap a page control to leave display input. Ctrl] is ordinary guest input; native mirror prefix commands do not apply here. For local builds/hosting, use the browser tooling instructions.

Browser compatibility and compact keyboards

Use a current desktop browser. Linux Chromium has automated integration coverage; we also reported Chromium, Chrome and Edge working in our earlier standalone setup. Those reports do not establish every browser/OS combination. On 261013 we also reported usable Android mobile operation; device/browser versions and a complete touch/keyboard matrix were not recorded. A normal HTTPS tab must allow service workers and the shared-memory facilities used at startup; a blocked facility produces a startup error.

macOS remains unqualified. On 261012 we tried an **Intel MacBook Air**: some Fn combinations worked, Alt Space did not, and application exit worked with Alt Esc rather than Alt Q. The browser, macOS version and exact Fn combinations were not recorded. This is a limited user observation, not a complete Mac keyboard test or a native macOS port.

Apple documents Fn Left / Fn Right for Home / End, Fn Up / Fn Down for PgUp / PgDn, and Fn Delete for forward Delete (the Delete navigation key); see Apple's keyboard reference. Their delivery with our additional modifiers still needs checking. In these guides Alt corresponds to Option on a Mac; Control is not Command. The browser keeps Command shortcuts, and Option-generated characters or OS shortcuts can prevent a chord reaching the terminal. No universal Insert substitute is assumed.

The reference route table identifies browser/native differences. On-screen navigation keys and command help are proposals; they are not currently present in the page.

Start a desktop or hardware session

For a fresh checkout or machine, first read the Build Guide.

Open your board's folder under `src/RunCPM_7xtgv-terminal/tools/targets/` on Linux, or `src/RunCPM_7xtgv-terminal/tools-win/targets/` on Windows. The desktop launcher names match: use `.sh` on Linux and `.cmd` on Windows. For example, on Linux from the repository root:

```
cd "src/RunCPM_7xtgv-terminal/tools/targets/EP4_10 -- jc8012p4a1c"
```

Use	Launch from the board folder
Desktop, local card	`./desktop-local.sh` uses the last desktop build and repository `cpmcard/`.
Desktop, mounted card	`./desktop-card.sh` asks for the mounted filesystem root containing `A/0`, or takes `--card-root PATH`.
Physical board	Connect the matching module and run `./mirror.sh`; this does not build or flash it.

On Windows, run `desktop-local.cmd` from Command Prompt or Explorer after `desktop-rebuild.cmd` succeeds. For an external card, use `desktop-card.cmd --card-root "E:\"`, replacing E: with the actual card drive. Windows keyboard input goes to Mintty; CP/M commands and INI settings are the same as on Linux. Hardware launch/upload commands use the Linux tools.

- Build the desktop first with `./desktop-rebuild.sh` if no executable exists or source has changed. Hardware rebuild/reflash are separate commands.
- Run the script in a terminal or the `.cmd` launcher from Command Prompt or Explorer. The shortcut already selects its board.
- On desktop, **type into the mirror**, not the output-only SDL window. Close either owning window, or use guest `EXIT`, to stop the emulator.
- Both desktop card modes are read/write, including INI/history autosaves. Do not run two writers on one card, and close before unmounting it.
- Hardware uses the shared 7xtgv-terminal mirror session (`7xtgv-terminal`); desktop uses its own isolated session. Never open a second serial reader alongside the hardware mirror.
- Ctrl], then q, exits the host bridge. With a desktop owning mirror, ending the session also stops its emulator; hardware CP/M keeps running.

Wait for the CP/M prompt. A brief mirror redraw after USB reconnect is normal.

- Automatic mirror selection prefers Konsole, then Kitty, then the current terminal; `--mirror terminal` selects the latter explicitly.

- For native Linux sessions, the shortcut tables assume the supplied Konsole/Kitty mappings. An arbitrary current terminal or desktop window manager may intercept keys.
- The older direct `kon.sh`/`kit.sh` entry points use host target/port configuration; details are in the Reference Guide.

First contact
power, boot and configuration

- Power the module from a suitable supply; large panels can overload a USB hub.
- A successful boot reaches `A0>` or a configured 7XF named prompt.
- `Build` identifies the firmware; `Module` identifies the panel and view.
- Hardware `Storage` shows the requested Arduino SPI clock, not a measured clock. Desktop instead reports its host-filesystem adapter and `Backend: desktop`.
- C6 additionally reports internal heap; two FreeRTOS tasks share one CPU.
- A red diagnostic or `CPU halted` is not a normal successful startup.

Know which configuration is active

- `A0:7XTGVTRM.INI` holds global settings, display settings and ten application profiles.
- `A0:7XTGVTRM.BUF` holds their histories; do not edit BUF as text.
- The card can move between modules: each display has its own INI section.
- Desktop reads these files inside the selected card root. It uses the emulated board’s display ID, not a separate SDL configuration section.

2. Everyday use

Profiles, history and application controls

Choose the application profile

- Alt 0 through Alt 9 select a profile; Alt Space shows its name.
- Alt PgUp / Alt PgDn browse previous / next profiles.
- Wait for the temporary name label to disappear before typing into the application.
- Selection may send the INI `on_select` command **at a live CCP prompt only**. It may change the DU; selecting a profile is not necessarily launching an app.
- Inside an application, selection changes controls without sending `on_select`.
- If keys seem wrong, check the selected name and choose the intended profile.



Windows: Alt Space opens the system window menu. This is a Windows shortcut, not a browser-only restriction; it can also intercept the chord in native terminal use. In our browser version, Ctrl Space is the alternate chord for showing the current profile. This alias is not currently provided by our native terminal keymaps.

Browser: Alt Ctrl PgUp / Alt Ctrl PgDn are fixed aliases for Ctrl PgUp / Ctrl PgDn when the browser captures those chords. They use the selected profile's existing application navigation bindings, without changes to Turbo or FAND. The browser aliases are available on all browser hosts.

Our installed application controls collect the Turbo/TINST map, MBASIC and BBC BASIC keys, Czech keyboard pictures and the unresolved FANDINST/Sharp adaptation. These are our working-card installations, not universal vendor defaults.

Everyday terminal shortcuts

PC key	Everyday effect
Ctrl Shift PgDn / Ctrl Shift PgUp	Zoom in / out temporarily.
Ctrl Shift with an arrow key	Pan the display viewport by axis.
Ctrl Shift Home / Ctrl Shift End	Scan all viewports backward / forward.
Ctrl Shift X	Native mirror only: open the transfer panel at an idle CCP prompt.
Ctrl Alt C	Force a running external COM back to CCP if it ignores Ctrl C; cancel the remaining SUBMIT batch.

The complete key table includes backlight, modes, exits, history deletion and host-layout notes. Normal zoom and pan do not edit the INI or resize the logical CP/M screen.

Emergency return is not a normal application exit

- Ctrl Alt C cancels the external COM and remaining SUBMIT commands.
- It is inert at CCP or inside an internal CCP command; ordinary Ctrl C is unchanged.
- This is an **unclean** return: unsaved application data and final writes may be lost.
- It cannot recover a wedged device or SD driver. Prefer the application's own exit.

Recall and edit history

Try	What happens
Up / Down	Recall history in line-oriented profiles; type a prefix to filter.
Alt Up / Alt Down	Recall history where the profile gives plain arrows to its editor.
Ctrl Shift Backspace	Delete the currently recalled unwanted entry.
Alt Backspace	Configured clear action: typically CLS at CCP, app-specific elsewhere.
Alt Q (Alt Esc in Konsole/Kitty or the browser; see the MacBook note)	Configured application exit: for example SYSTEM in MBASIC, QUIT in BBC BASIC.

- Application settings come from each complete `[bankN]` INI section. To reorder profiles, move the whole section and change its number. Slots share neutral fallbacks; names do not activate hidden application settings. If moving the CP/M profile, update `[global] ccp_bank` too.
- Each of ten profiles keeps up to 100 history entries.
- History retains the first 79 characters per entry (80 with the saved `>` marker). Longer active input is still supported; older longer history entries are shortened when loaded and remain shortened on the next save.
- At a recognized CCP prompt, the INI's `ccp_bank` handles shell commands even while another profile is selected; the working card assigns bank 0. Without that setting, automatic routing is disabled.
- Enter saves the line you submit, including filtered recalls and completed numbered lines. Browsing without submitting does not save another entry.
- Empty/single-character lines and exact adjacent repeats are not appended. Reusing an older command makes it recent again; spaces remain significant.
- Separate editable keyword lists are a plan, not a delivered feature.

Temporarily bypass profile input

- Alt Shift 3 toggles `## RAW` direct-input mode.

- Use it when an application needs input without the normal profile/history handling.
- Toggle it back for ordinary command work. Direct Alt 0 through Alt 9 selection also leaves RAW mode; profile browsing does not.

Explore CP/M with the 7XF shell

`7XF` is an internal CCP command, not a COM file or a separate filesystem. Names come from catalogs; CP/M still uses flat drive/user areas.

```
7XF
LS /
LL /A
CD /B/MBASIC
LS
LL MBASIC.*
DIR MBASIC.*
LDIR MBASIC.*
CD /A/CPM
CD /B
```

- `LS` lists short sizes/names; `LL` or `LS -L` adds details.
- File lists sort by extension, then name; displayed filenames use lowercase `name.ext`. Files on the card are not renamed.
- `LS ..` / `LL ..` lists the current drive's areas; `LS /` lists all drives. Both show the whole-drive heading. `LS ..` omits individual area descriptions; `LL ..` includes them. Root `LS /` and `LL /` retain drive descriptions. Empty/catalog-only areas are hidden from area overviews, not deleted or made inaccessible. The current-area file heading shows just its description from column 2, without repeating the prompt's path.
- Descriptions stay beside their rows. A wider logical terminal shows more text; small widths clip descriptions rather than adding a second line.
- `CD /C/xterm256` selects `xterm256(C2)`; aliases depend on your card.
- `7XF DD` returns to decimal `A0>`; `7XF DH` uses hexadecimal users.
- `7XF D` shows only the drive; `7XF DN` selects the named prompt.
- `VOL` remains the original INFO.TXT command; it does not use 7XF catalogs.

- `DIR` and `LDIR` still work from named prompts: traditional CP/M listings, without 7XF descriptions or its extension sorting. `LDIR` adds sizes.
- `/` is a catalog overview, not a real root directory. `CD /B` returns to the remembered B area; it does not enter a hierarchical folder.
- `7XF` selects the named prompt only; catalog commands already work from `A0>`. For named prompts on boot, a card-root `AUTOEXEC.TXT` can contain `7XF`. This is separate from INI; do not overwrite an existing startup command.
- An INI profile can use `on_select=cd /b/mbasic\r` as a named-area shortcut when selected at a live CCP prompt. It is not a boot command and is not injected into a running application. Names, banks and bindings are user choices.

Missing aliases do not mean files disappeared. Use the ordinary DU, e.g. `C2:`, or inspect the actual catalog. Our repository and your card can differ.

Check or change terminal size

Run these commands at the CP/M prompt:

Command	Effect
`VTINFO`	Show display ID, target, build, panel pixels, logical size, viewport and presets.
`VTZOOM`	Show the current view.
`VTZOOM 2`	Temporarily load preset 2, including its logical terminal size.
`VTZOOM 2 80x24`	Use an 80x24 logical screen at 2x zoom without saving it.
`VTZOOM DEFAULT`	Restore the INI/HAL boot view.

Match the application before launching it

- Ordinary keyboard zoom changes physical scale, not the logical screen size.
- A full-screen app installed for 80x24 expects that logical geometry; panning cannot correct a mismatch in its cursor positioning.

- Select the application profile first, then set the intended geometry before launching the app. Profile selection can also apply per-profile geometry.
- On a small panel, an 80x24 terminal occupies several panned views; the native mirror still shows all 80x24 cells. Browser **Screen text** provides a frozen full-screen copy.

See the geometry rules.

Colors, emphasis and brightness

Palette and text emphasis

- Try `VTPAL 0` through `VTPAL 5`: PAPER, green CRT, amber CRT, white CRT, red plasma and cyan VFD. INI edits can redefine these presets.
- Palette selection is saved lazily; `SYNC` requests pending persistence.
- On light paper, emphasis can be darker; on dark paper it can be brighter.
- ANSI colors map to the terminal's own ink/paper roles; this is not full ANSI.
- CLS clears cells without resetting text attributes.
- If Turbo leaves CCP emphasized, configure TINST's **terminal reset** string as `ESC [ 0 m`. Initialization and exit reset are different settings.

Panel backlight

Key	Backlight
Alt Ctrl Shift PgUp / Alt Ctrl Shift PgDn	Raise / lower through the module's tuned levels.
Alt Ctrl Shift 0	OFF; raise brightness to restore it.
Ctrl Shift 1 through Ctrl Shift 9	Still-supported coarse presets; fine stepping is preferable.

- Backlight affects the panel, not PC mirror colours; the last non-off level persists.
- From OFF, the first step in either direction restores that remembered level.
- EC6_147 is capped at 50% duty following vendor thermal advice.

- Equal level numbers need not mean equal brightness on different panels.
- SDL and browser brightness simulate the panel; they do not establish physical light output.

Select the terminal output mode

- ANSI/VT100 is the usual output mode.
- `VT52` / `VT100`, or Alt Ctrl Shift 2 / Alt Ctrl Shift 1, select the implemented output interpretations.
- RAW input, zoom, profiles and graphics are separate controls.
- ADM3a and the full native 7xtgv chip specification remain future work.

3. Files and settings

Transfer one file

These steps use the **native mirror** on desktop or physical hardware. The standalone browser has no transfer panel; use card ZIPs there.

1. Leave CP/M idle at its prompt in the **guest/device** DU you want.
2. Open the panel with Ctrl Shift X.
3. Choose S (host to CP/M) or R (CP/M to host).
4. Select the **local** DU folder, source name and target name.
5. Review the direction and destination before confirming a write with Y.

Two concrete examples

- **Send VTESTS:** select the intended test DU on the device, choose S, local `C1:` (`7xtgv(C1)`), source `VTESTS.BAS`, target `VTESTS.BAS`.
- **Receive into a different local area:** leave the device at `A0>`, choose R, then local `A3:` and the file names.
- The panel never switches the guest/device DU; its local DU is a host folder.

Safeguards and recovery

- Sending always asks before writing the device target; receiving asks before replacing an existing local file. Enter alone does not approve replacement.
- Bytes, including CP/M record padding, are preserved; there is no newline or character-set conversion.
- Use an 80-column **logical** terminal. A narrower panned viewport is fine.
- After a stalled helper, return to a clean CCP prompt or reset before retrying.
- On desktop, the selected card already is a host folder. For direct host-file copies or edits, close the emulator first rather than racing its card access.

Exact options and failure behaviour are in the transfer reference.

Keep or replace a browser card

Save and exit the guest application, click **Stop**, and wait for **Stopped cleanly**. **Download card ZIP** saves the memory card with a top-level `cpmcard/` folder. After stopping, **Load card ZIP** replaces the current card and starts it; **Restart current card** instead resumes from the stopped card in a fresh runtime. Download wanted changes before loading a different ZIP.

Closing/reloading the page discards unexported changes, including INI and history saves. Guest `REBOOT` and target switching preserve card contents but restart execution. The public site does not receive uploaded cards or keep an automatic backup. ZIP files are independent snapshots, not synchronization with a native host directory or physical SD card.

To edit browser-card files on the host, export the stopped card, keep a backup, extract and edit the intended files, then ZIP the `cpmcard/` directory and load it back. Preserve CP/M text/binary formats and filenames. The browser accepts older ZIPs rooted directly at `A/0` too; limits and validation are in the browser tooling reference.

Edit the selected card's INI safely

Desktop or directly mounted card

1. Close the emulator before editing its card files.
2. Back up `cpmcard/A/0/7XTGVTRM.INI`, or that path under your selected card root.
3. Edit only the intended settings, save, then relaunch.

Do not fetch from an unrelated physical board over your local settings. These steps also apply to a mounted card used directly by desktop.

Physical module
fetch, edit, send, reboot

1. Fetch its **live** A0 file using `tools/fetch-7xtgvtrm-ini.sh` from the sketch directory; specify target/port if needed.

2. The helper archives the previous local copy and replaces it only after a successful fetch. Stop if fetching fails: the repository copy may be older.
3. Edit the fetched file, preserving unrelated settings.
4. Send it back to device A0 with explicit overwrite approval.
5. Reboot to apply settings loaded only at startup.

Alternative
edit inside CP/M

- Choose the TURBO profile and open `A0:7XTGVTRM.INI` in its configured editor.
- Verify the file's area, save, exit and reboot; do not interrupt a save.
- Avoid editing a similarly named copy in another DU. TE is unsuitable for some long INI files.

Example settings fragment

```
[global]
sdfat_spi=20

[display:EC6_147]
view_zoom=1
zoom1_terminal=80x24
zoom2_terminal=80x24
```

- This is a **fragment**, not a replacement INI; retain the other settings and profiles.
- View fields describe boot presets; ordinary keyboard zoom does not rewrite them.
- The full grammar, diagnostics and persistence rules are in the INI reference.

4. Troubleshooting and checks

Use the recovery checklist first. The diagnostic commands and optional demonstrations below help check the current module and terminal behavior.

When something looks wrong

- Browser ignores typing: click the display or **Keyboard → display**. A browser or OS may reserve a shortcut; see compatibility notes.
- Browser startup unavailable: open the published HTTPS page in a normal current browser tab and read the displayed error. No local compiler is needed; hosting requirements are in the browser tooling README.
- Browser changes disappeared after reload: only an exported ZIP can restore the previous memory card. Guest saves alone do not survive page closure.
- Browser display too large: reduce **Display** scale. To see more terminal cells, change glyph zoom/pan instead; **Screen text** shows the complete text grid.
- No hardware mirror or wrong board: recheck the selected target folder, serial port and cable; stop a stale shared session before relaunching. Changing the host target does not identify the attached hardware automatically.
- SDL receives focus but not typing: select the mirror. Always-on-top does not prevent the SDL window from taking focus through a click or Alt Tab.
- RP2350 firmware recovery: enter BOOTSEL manually for a requested flash. Do not copy another target's UF2. Flashing is a separate operation from CP/M file transfer.
- `LOADED WITH ERRORS`: inspect boot diagnostics and `A0:7XTGVTRM.ERR`; use the INI diagnostics before editing the live file.
- Distorted full-screen app: compare `VTINFO` logical size with the application's installed size. Physical zoom alone will not fix a mismatch.
- Transfer panel/helper stuck: return to a clean CCP prompt or reset, restore an 80-column logical terminal, and retry. The transfer normally restores the original profile and RAW state.

For source-level diagnosis and flash safeguards, use the Technical Reference Guide.

Inspect and compare the module

Command	Use
VTINFO	Module, build, logical screen, viewport and zoom presets.
VTCPU	Compare emulator speed with terminal active and quiet.
SPEED 65535 / SPEED 0	Use the fastest throttled CPU setting / restore unthrottled execution.
VTTERM	Run about three seconds of coloured text/parser and display redraw work; report kChars/s and kB/s (M above the million threshold).
VTDRV	Read/verify its existing 32 KiB test fixture.
VTDRV W	Create/write the fixture, then read/verify; writes the current DU.
VTDRV W Q	Quiet timed write/read passes; matching firmware required.
CARDMARK 4096 40 CSD	Hardware-only raw reads and decoded card registers; no test-file writes. Unsupported on native desktop and browser.
SYNC	Request pending history/config saves.
REBOOT	Restart the module, native desktop process/mirror, or browser runtime. The browser keeps its memory card; this is not merely CP/M warm boot.
REFLASH	Hardware: request the configured host uploader; cannot build firmware itself. Native desktop and standalone browser cannot invoke it.

CARDMARK clocks are requests, not measured SCK. Read agreement is not write qualification; use a backed-up/spare card for experimental clocks.

For a quick display comparison, run `VTTERM` at the CP/M prompt. It clears the screen and scatters NORMAL/ALT coloured text on random paper slots 0..3 using the normal terminal render loop. Compare the same target, logical geometry, glyph zoom and output/mirror setup. Elapsed time is in integer milliseconds without grouping. Total characters and rates use decimal thousands (`k` = 1,000) or millions (`M` = 1,000,000), with three decimal places; this measures native terminal work, not USB link throughput. See VTTERM measurement details and source location.

Try the MBASIC terminal tests

Select the MBASIC profile first, then select `7xtgv(C1)` below. The full language package lives in `mbasic(B1)`; our terminal tests live in `7xtgv(C1)`. Use 80x24 logical geometry for the scroll-region checks:

```
C1:
VTZOOM 1 80x24
MBASIC
LOAD "VTESTS"
RUN
```

- In the menu, try 3 for attributes, 4 for DEC graphics, 5 for VT52, or 9 for scrolling regions. Follow the on-screen prompts; Q returns to CP/M.
- Scroll regions span complete logical rows, not arbitrary rectangular windows.
- Also try `LOAD "VTPAL"`, then `RUN`, from the BASIC prompt.
- Alt Q uses the configured SYSTEM exit; Alt Esc is also available in Konsole/Kitty and the browser. See the MacBook observation.
- Save maintained BASIC with `SAVE "NAME",A` for ASCII; ordinary SAVE can create tokenized data.
- Back at CCP, `VTZOOM DEFAULT` restores the boot view.

Try the xterm256 plot

Use `7XF`, then `CD /C/xterm256`, or select `xterm256(C2)` by DU:

```
C2:
CLS
SUBMIT BROT_BS
```

- `SUBMIT BROT_PS` is the alternate renderer; both scripts initialize the plot.
- The card's scripts choose fixed pixel positions. A small panel may clip the image; choose EP4_10 in the browser at Display 1x or use its native desktop for a spacious demonstration. A narrow browser window may still need scrolling.
- Ctrl Alt C can abort a running COM and cancel the remaining script.
- Completion or forced return ends graphics mode, retaining the image until later text/repaint replaces it.
- The pixel plot appears on the TFT, SDL panel or browser canvas. The native text mirror and browser **Screen text** do not contain those pixels; use PNG capture in the browser to keep the image.